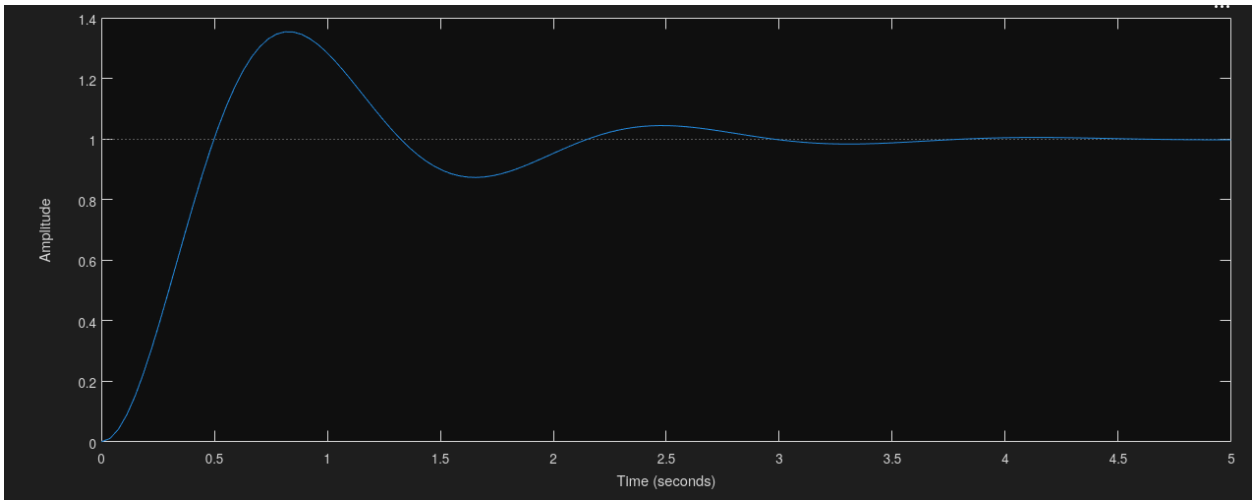
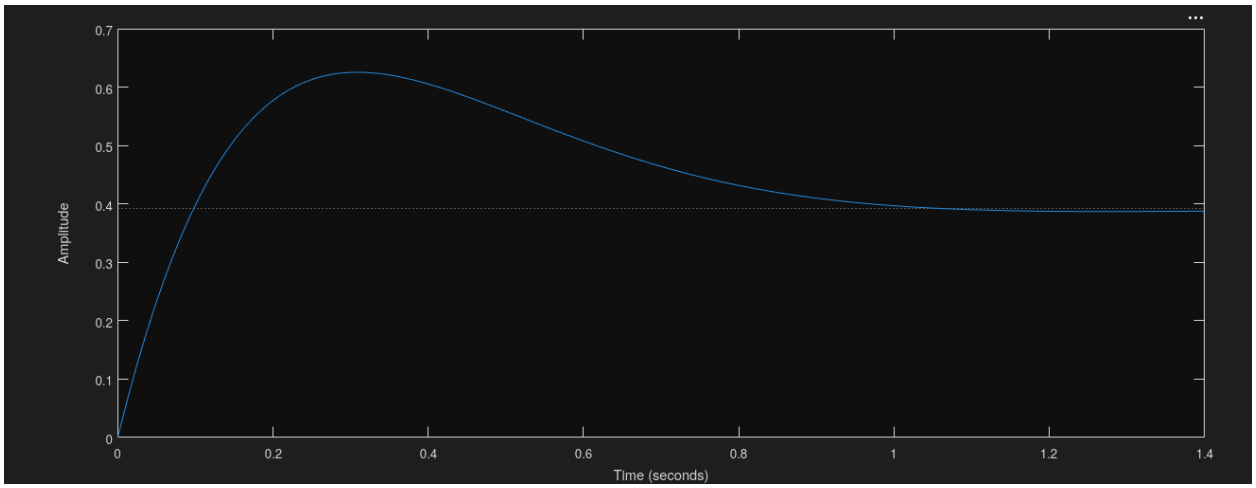
1.

B.



C. SettlingTime: 2.798
Overshoot: 35.4991

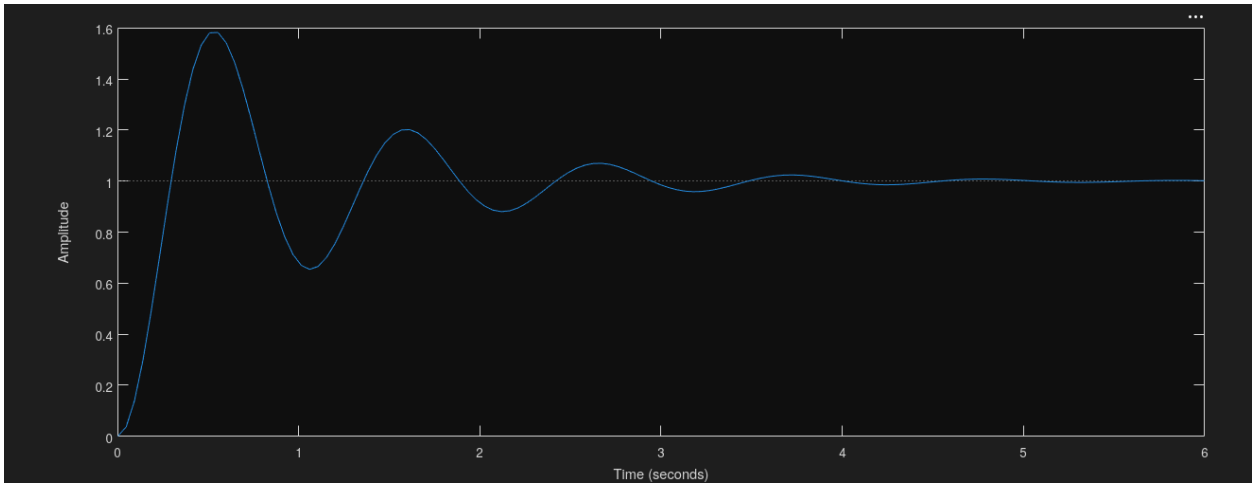
F.



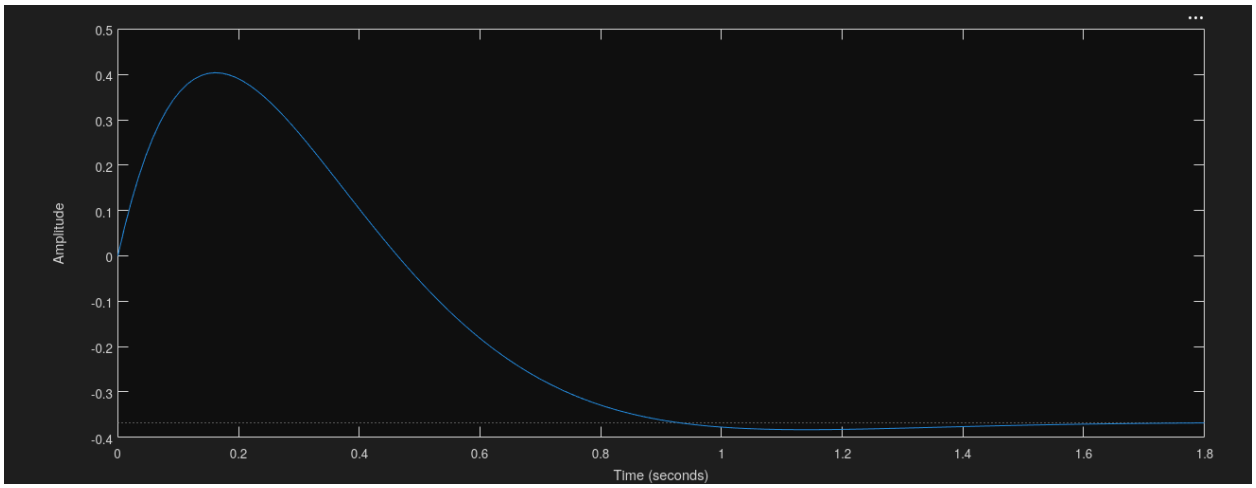
Theres a huge over shoot and I am not sure why

2.

B.


Amplitude
Time (seconds)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
0
1
2
3
4
5
6

F.


Amplitude
Time (seconds)
-0.4
-0.3
-0.2
-0.1
0
0.1
0.2
0.3
0.4
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8

3.

C)

```
>> [A,B,C,D]=tf2ss(NUM,DEN)
A =

   -1.8000

B =

     1

C =

    0.6000

D =

     0

>>
```

4.

c)

```
>> [A,B,C,D]=tf2ss(NUM,DEN)

A =

   -4.0000   -7.5000
    1.0000         0


B =

     1
     0


C =

    1.5000    1.0000


D =

     0
```

5.

```
>> [A,B,C,D]=tf2ss(NUM,DEN)

A =

   -2.6667  -10.0000  -33.3333
    1.0000         0         0
         0    1.0000         0


B =

     1
     0
     0


C =

    0.3333    0.6667    0.3333


D =

     0
```

6.

```
>> [A,B,C,D]=tf2ss(NUM,DEN)

A =

    -8   -30
     1     0


B =

     1
     0


C =

    -1   -20


D =

     1

>>
```

7.

```
>> [A,B,C,D]=tf2ss(NUM,DEN)

A =

    -7   -20
     1     0


B =

     1
     0


C =

   -22   -70

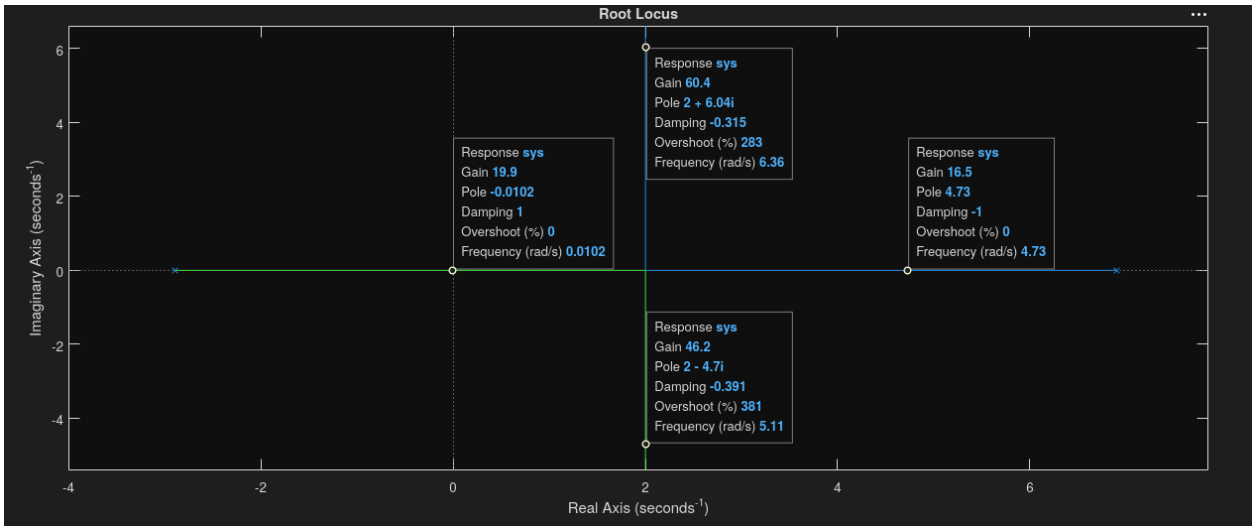

D =

     4

>>
```

11.

e)



H)

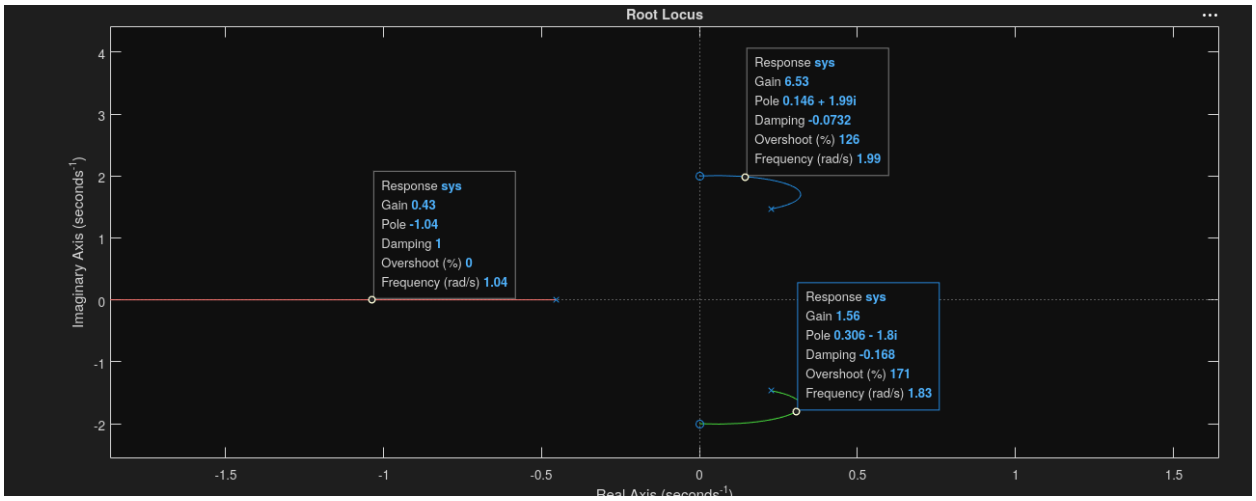
```
>> pole(sys)

ans =

    6.8990
   -2.8990
```

12.

e)

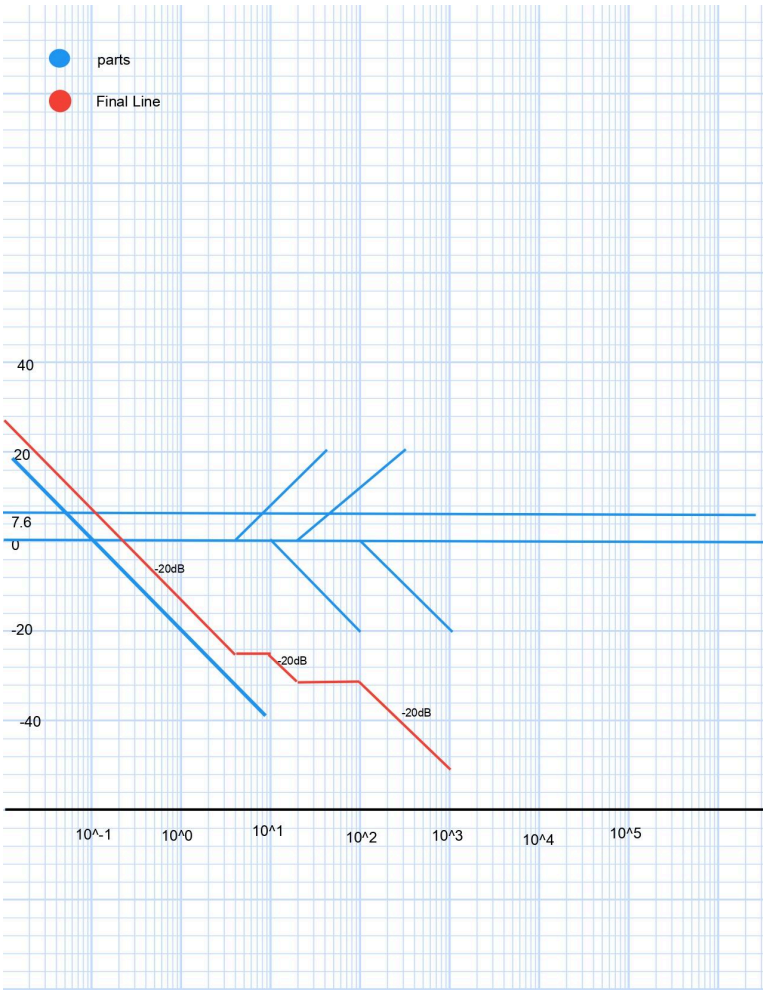


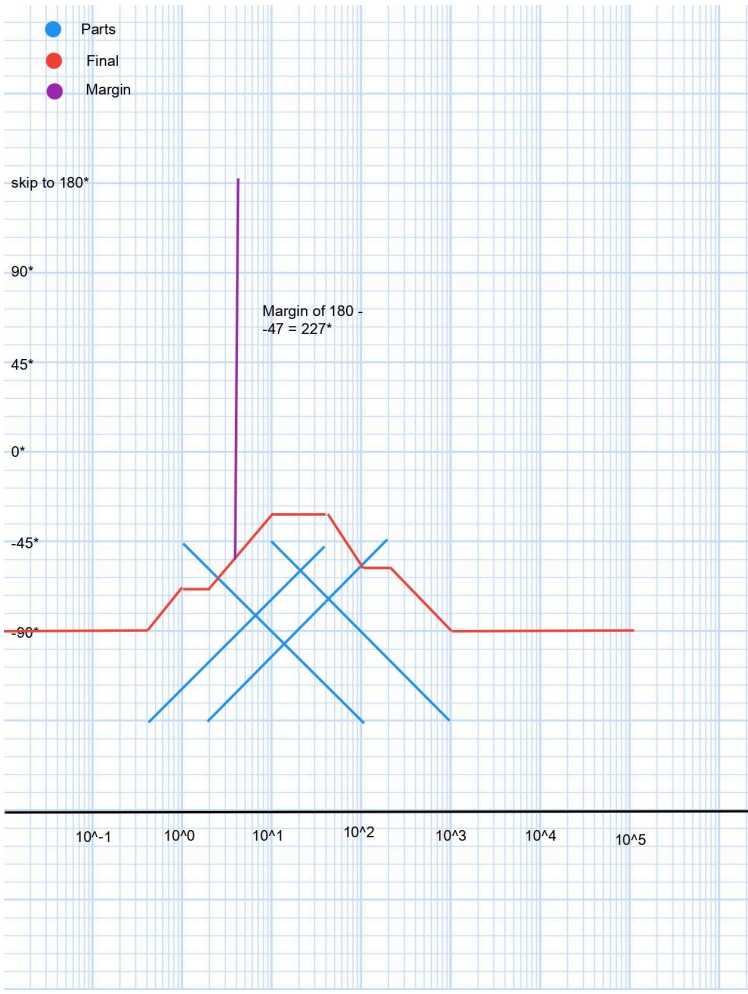
H)

```
ans =

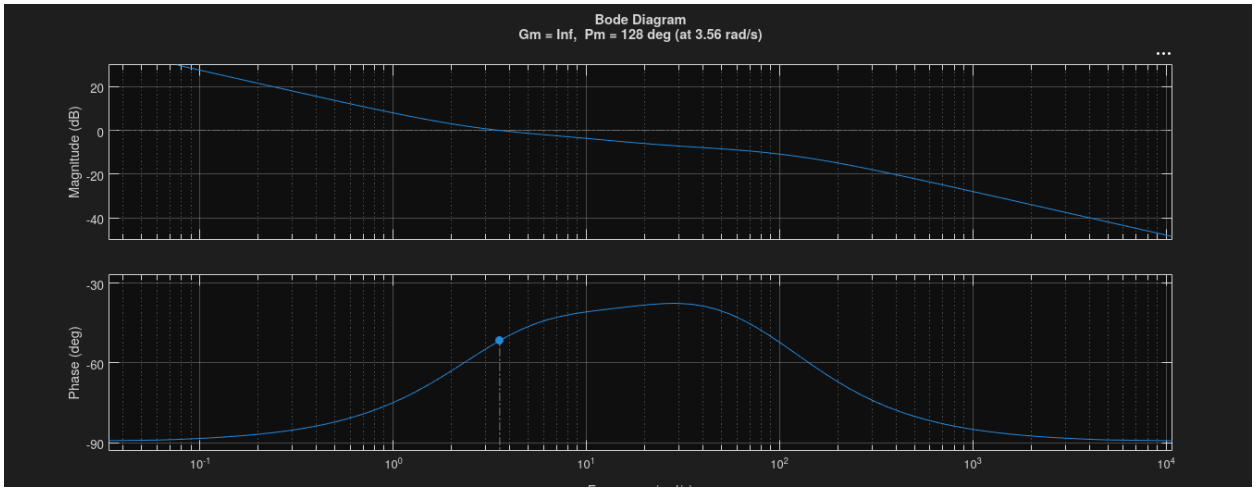
  -0.2151 + 1.3071i
  -0.2151 - 1.3071i
  -0.5698 + 0.0000i
```

13. Phase plot never reaches 180 degrees





f)

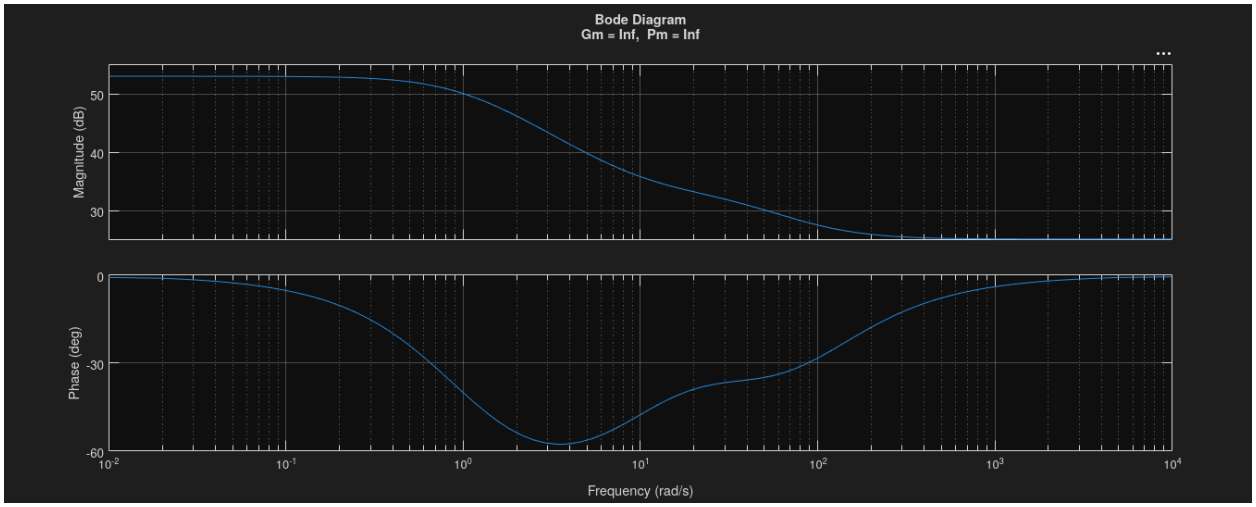


g) Gain margin is infinite and Phase margin is 128. (I am now realizing i was supposed to find the difference between -52 and -180)

H) The system is unstable because PM is less than GM

14.

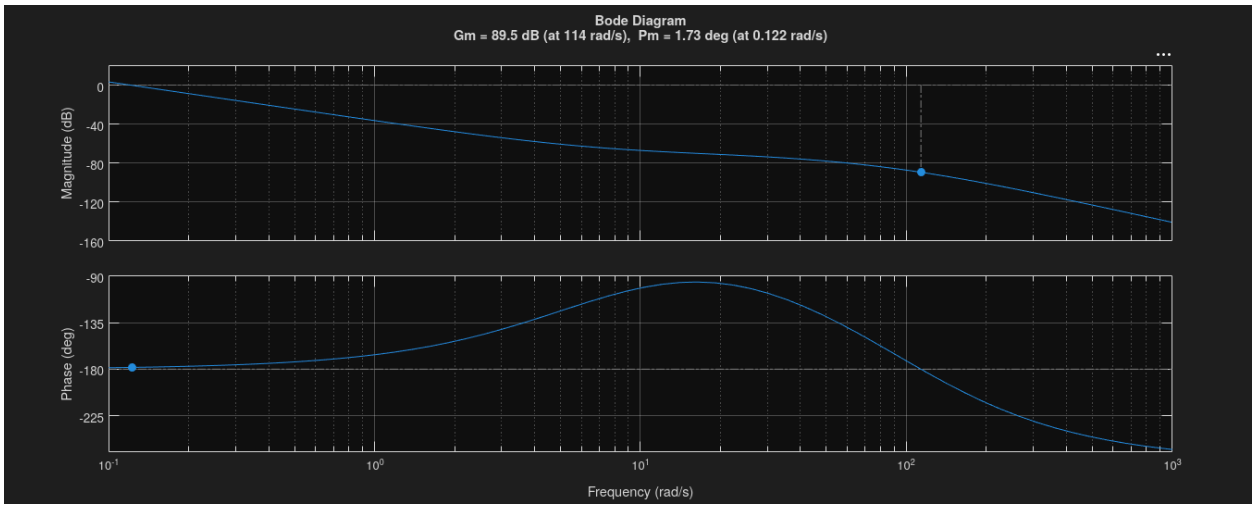
f)



g) the margins are infinite

h) this is marginally stable since GM and PM are technically equal

15.
f)

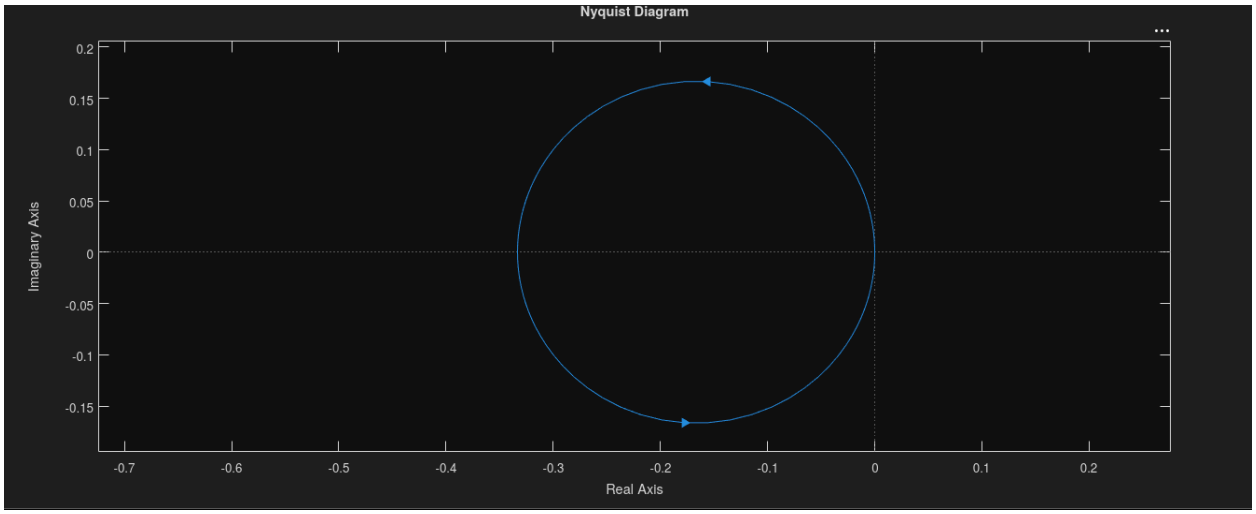


g)
GM: 89.5 PM:1.73 degrees

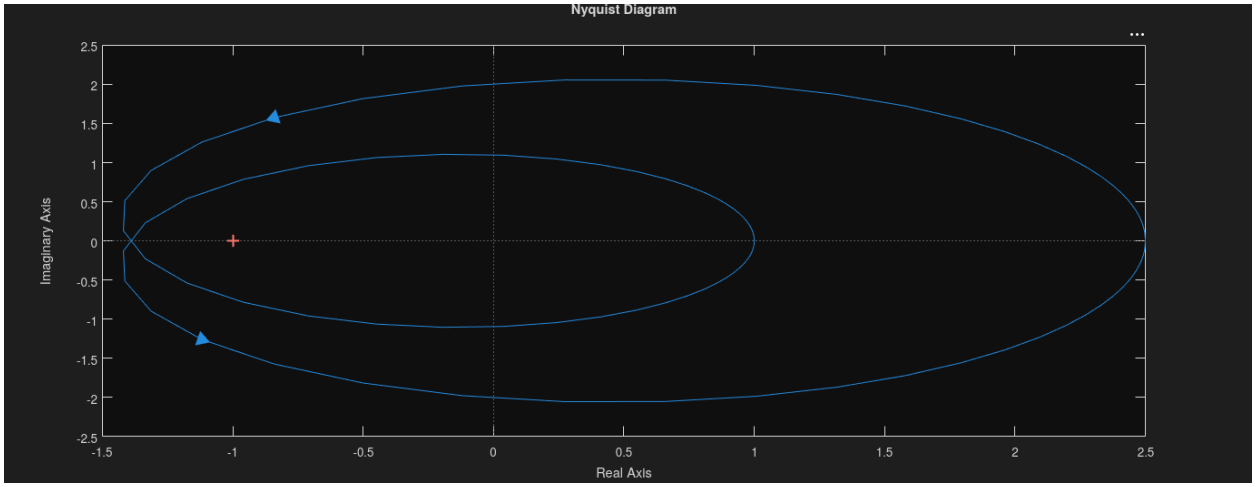
H)
The system is stable because GM > PM and both are positive

16.

e)


Nyquist Diagram
Imaginary Axis
Real Axis
0.2
0.15
0.1
0.05
0
-0.05
-0.1
-0.15
-0.7
-0.6
-0.5
-0.4
-0.3
-0.2
-0.1
0
0.1
0.2

17.


Nyquist Diagram
Imaginary Axis
Real Axis
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
-1.5
-1
-0.5
0
0.5
1
1.5
2
2.5